Mechanism Is One Nonlinearity Upstream: Exact Program Recovery from Trained Networks, and Where Behavioural Distillation Fails

Jabir Hussain
Independent researcher
jabir01hussain@gmail.com

Preprint, October 2026

Abstract

Early mechanistic interpretability reverse-engineered small networks by hand into explicit programs, and then largely stopped: the method was expensive, not refuted. This paper tests whether an automated engine can reprice it, under a sealed pre-registration fixing success criteria, kill conditions, budgets and seeds before any code was written. An SMC-style search over a typed program space, scored exactly against the network (behavioural fit, interchange interventions, and a description-length prior), recovers a program exactly equivalent to the grokked modular-addition network of Nanda et al. (2023) on 5/5 declared seeds, in about twenty seconds per seed. It recovers the network's *mechanism* on 0/5, and the reason is proved. If the network's outputs leak mean probability mass $\bar{\varepsilon}$ off the argmax, then any two argmax-correct candidates differ in total distributional log-likelihood by at most an explicit $O(N\bar{\varepsilon}\log(1/\bar{\varepsilon}))$ bound (Proposition 1); on the grokked network this bound is 0.283 nats against a description-length gap of 9.525 nats, so the prior provably selects the shorter, mechanism-free program (Corollary 1). The information is not destroyed, only hidden: one nonlinearity upstream, the published mechanism explains 91.6% of the network's logit variance against 8.2% for a lookup, and under a profile Gaussian likelihood on logits the preference inverts decisively (Proposition 2), robustly to a 866-fold misspecification of the noise scale. The evaluation programme is released as an exactly graded program-recovery benchmark: 30 procedurally generated Tracr-compiled targets across a 57-fold parameter range, scored by execution over the full input space with a pre-registered calibration protocol. The engine is its first strong baseline: it fails the pre-registered recovery bar (5/20 small, 3/20 medium, 3/20 large tier) yet is the only method among four that recovers anything beyond the smallest tier, and its stated confidence is informative (Spearman $\rho = 0.462$, bootstrap 95% CI $[0.251, 0.629]$) while being badly overconfident (ECE = 0.629). Four defects found in the experimental instruments, two of which flattered the method, are reported alongside the results.

Keywords: mechanistic interpretability; program synthesis; distillation; minimum description length; benchmarks; pre-registration.

1 Introduction

The earliest convincing results in mechanistic interpretability were obtained by hand: a researcher stared at the weights of a small trained network until the computation could be written down as an explicit program, and the program was then checked against the network (Olah et al., 2020; Elhage et al., 2021; Nanda et al., 2023). The method produced accounts one could actually falsify. It was then largely abandoned, and the stated reason was never that the accounts were wrong. It was cost: one small network could absorb weeks of expert attention, and the field moved to cheaper methods that trade completeness for coverage. This paper asks a narrow, decidable question: can an automated search,

with no human in the loop, recover program-level accounts of small networks, and does it know when it has failed?

The answer procedure was fixed before anything was built. A sealed pre-registration, written before any code, fixed the success criteria, three kill conditions with numeric thresholds, the scoring function, the seeds, and the budgets; it has not been edited since, and its one amendment is published with the algebra that forced it. Both outcomes of every test were acceptable in advance. The result is a split verdict, and the most useful finding is the one the study did not set out to find.

Contributions.

1. **The softmax ceiling** (Section 4). It is proved that, for a network whose output distributions concentrate within mean leaked mass $\bar{\varepsilon}$ of a point mass, the distributional likelihood cannot separate any two argmax-correct candidate programs by more than an explicit bound that vanishes as $\bar{\varepsilon}\log(1/\bar{\varepsilon})$ (Proposition 1); whenever a description-length prior gap exceeds that bound, maximum a posteriori selection provably prefers the shorter program *whatever its mechanism* (Corollary 1). On the grokked modular-addition network the bound evaluates to 0.283 nats against a prior gap of 9.525 nats: the published Fourier mechanism of Nanda et al. (2023), hand-written into the engine's own candidate language, fits the outputs marginally *better* than a one-hot lookup and loses the selection. The realised likelihood gap, 9.46×10^{-5} nats, is three orders of magnitude below even the bound.
2. **The inversion, one nonlinearity upstream** (Section 5). The mechanism's signature survives in the pre-softmax logits: it explains 91.6% of the network's logit variance (lookup: 8.2%), with mean correlation 0.956 on the coordinates below the argmax, where the lookup is degenerate by construction. Under a profile Gaussian likelihood on logits the model-selection inequality reverses by five orders of magnitude (Proposition 2), and the reversal is robust until the assumed noise scale exceeds the observed residual by a factor of 866. To the author's knowledge, prior work reads pre-softmax structure for analysis (nostalgebraist, 2020); placing a *distillation likelihood* there, and quantifying the preference inversion it buys, is new.
3. **An exactly graded program-recovery benchmark and its first baseline** (Sections 6–7). Thirty procedurally generated RASP programs compiled to transformers with Tracr (Lindner et al., 2023), novel by construction, in three size tiers spanning a 57-fold parameter range, graded by exact behavioural equivalence executed over the full input space, with secondary axes for search-versus-selection failure and a pre-registered calibration protocol. The engine fails the pre-registered recovery bar and is simultaneously the only one of four methods that recovers anything beyond the smallest tier; both facts are reported with equal weight, together with a complete defect ledger for the instruments.

Everything in the paper regenerates from logged artefacts by one offline command, including a bit-identical replay of a recorded search (Appendix D).

2 Related work

Hand reverse-engineering and its lineage. Olah et al. (2020) articulated circuits as an object of study; Elhage et al. (2021) gave the transformer-specific framework. The grokked modular-addition network (Power et al., 2022) was fully reverse engineered by Nanda et al. (2023), who identified a Fourier multiplication algorithm and verified it with progress measures; it is the canonical example of a complete, falsifiable, and expensive account, and it is the Stage 1 target precisely because the ground truth is settled.

Automated circuit discovery and ground-truth evaluation. ACDC (Conmy et al., 2023) automates the localisation of circuit subgraphs; InterpBench (Gupta et al., 2024) supplies semi-synthetic transformers with known circuits for evaluating such methods, and Tracr (Lindner et al., 2023) compiles

RASP programs (Weiss et al., 2021) into transformers exactly. The benchmark differs in the object graded: these evaluations score the identification of *components*; this benchmark grades recovery of an executable *program*, by execution over the whole input space, which is a stricter and complementary standard. Circuit-level methods are not run here and make no comparative claim about them (Section 8).

Causal abstraction and multiple realisability. That behavioural equivalence underdetermines mechanism is the multiple realisability observation made precise by the causal abstraction programme (Geiger et al., 2021, 2025): distinct internal organisations can implement the same input-output map, and distinguishing them requires interventions on internals. The contribution of this paper to that line is quantitative: an explicit bound on how little the *output distributions* of a converged network can ever say about mechanism (Proposition 1), a measurement of where the lost information resides, and the observation that a description-length prior then actively inverts the preference rather than leaving it undetermined.

Program synthesis and distillation. DreamCoder (Ellis et al., 2021) established library-based Bayesian program search; symbolic distillation of learned models into formulae was demonstrated by Cranmer et al. (2020); knowledge distillation (Hinton et al., 2015) matches output distributions by construction and therefore sits squarely under the ceiling proved here. The scoring is a minimum description length posterior (Rissanen, 1978; Grünwald, 2007; Solomonoff, 1964), explored by a sequential Monte Carlo scheme (Del Moral et al., 2006); the term SMC-style is used advisedly, since the mutation kernels do not leave intermediate distributions invariant and no consistency guarantees are claimed.

3 Setup

3.1 Targets, candidates, and scoring

Definition 1 (Target)**.** A *target* is a finite input space $\mathcal{X}$ with $N = |\mathcal{X}|$, a network $f : \mathcal{X} \to \mathbb{R}^C$ producing logits over C classes (sequence targets produce logits per position; the position index is suppressed), the induced distributions $q_x = \mathrm{sm}(f(x))$, argmaxes a_x, and a fixed, seeded set of interchange interventions. The engine sees input-output behaviour and the intervention tables only: no weights, no architecture, no task name.

Definition 2 (Candidates and prior)**.** A *candidate* π is a program in a declared, versioned primitive language (arithmetic, modular and trigonometric operations, aggregation; sequence targets add selector and aggregate primitives), executed under an AST whitelist with no access to the network. $z_x^\pi \in \mathbb{R}^C$ denotes its logits on x. Its description length $\mathrm{DL}(\pi)$ in bits is a fixed prefix code over the syntax tree, and the prior is $\Pr(\pi) \propto e^{-\lambda \mathrm{DL}(\pi)}$ with $\lambda = 0.05$ nats per bit, fixed before any run.

Definition 3 (Distributional scoring)**.** The pre-registered data term is the temperature-calibrated distributional fit

$$\ell_{\mathrm{dist}}(\pi) \;=\; -N \cdot \inf_{\tau>0} \overline{\mathrm{KL}}_\tau(\pi), \qquad \overline{\mathrm{KL}}_\tau(\pi) = \frac{1}{N} \sum_{x \in \mathcal{X}} \mathrm{KL}\big(q_x \,\big\|\, \mathrm{sm}(\tau z_x^\pi)\big),$$

the scalar τ absorbing the candidate's arbitrary logit scale (a scalar offset is a softmax invariant). Causal alignment under input-interchange interventions enters with the same fitted τ; a candidate's log-posterior is $-\lambda\,\mathrm{DL}(\pi) + \ell(\pi)$ up to a constant.

Definition 4 (Recovery, and failure taxonomy)**.** A run *recovers* a target iff its top-posterior candidate is argmax-equivalent to the reference on every point of $\mathcal{X}$, checked by execution. A run is a *search failure* if no evaluated candidate was exactly equivalent, and a *selection failure* if one was and the posterior preferred another.

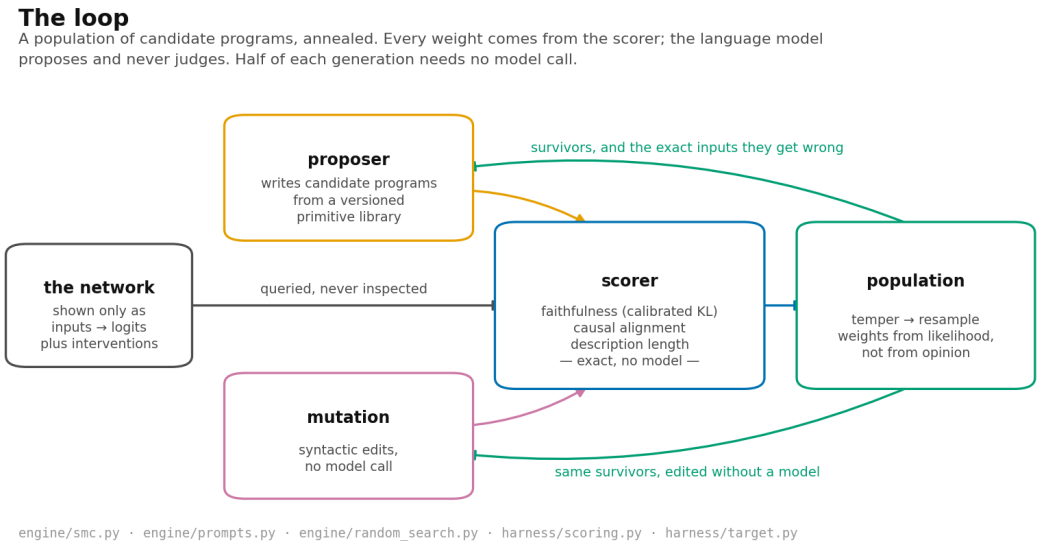


Figure 1: The search loop. Weights come only from the exact scorer; the language model proposes and never judges. Half of each generation needs no model call.

Definition 5 (Stated confidence)**.** At termination, candidates are grouped into behavioural equivalence classes by exact equality of predicted outputs over the evaluation set; the engine's confidence is the posterior mass of the top class within the final population. A secondary, post-hoc measure discounts this mass by a Good–Turing estimate (Good, 1953) of undiscovered competitive mass; it was declared before the Stage 2 runs and is reported losing (Section 7.2).

3.2 The engine, in brief

The engine is a tempered population search (Figure 1): a proposer LLM (DeepSeek-V4-Flash) emits candidate programs from the declared primitives; the harness scores them exactly per Definition 3; the population is reweighted under an adaptive temperature schedule targeting a fixed effective sample size, resampled systematically, and mutated, half by syntactic edits that need no model call and half by the proposer shown the survivors and the specific inputs they get wrong. A stronger model is called only after stagnation. The proposer proposes and never judges; prompts are linted against a banned-term list (no interpretability vocabulary, no task names) and a run aborts rather than send a contaminated prompt. All 105 proposer outputs in Stage 1 were scanned for naming the known solution unprompted: zero were flagged.

3.3 Pre-registration

The sealed document fixed: Stage 1 success (exact equivalence on the grokked network, five declared seeds, fixed compute and wall-clock caps), a secondary mechanistic criterion (the recovered program computes its logits as a small sum of periodic terms, tested behaviourally), Stage 2 pass thresholds (recovery $\geq$ 50% at the smallest tier and $\geq$ 25% at the largest, confidence positively rank-correlated with recovery), and three kill conditions: K1, failure on all Stage 1 seeds; K2, recovery collapsing to $\leq$5% at the largest tier while $\geq$50% at the smallest; K3, stated confidence uncorrelated with recovery (closed-form thresholds in Appendix B). Scoring coefficients were tuned only on development targets disjoint from every reported result and frozen before Stage 1. One amendment (a Stage 0 instrumentation criterion whose threshold was algebraically unsatisfiable) is published with its proof; it gates nothing downstream.

candidate	exact match	$\inf_\tau \overline{\mathrm{KL}}$ (nats)	DL (bits)	log-posterior gap
one-hot lookup	1.0000	1.05×10^{-6}	106.9	0
published mechanism	1.0000	1.04×10^{-6}	297.4	−9.525

Table 1: The inversion. Both candidates are argmax-exact on all 12,769 inputs; the mechanism fits the distributions slightly better and is rejected by the description-length prior. The network's mean output entropy is 3.97×10^{-6} nats against $\ln 113 = 4.727$.

4 The softmax ceiling

4.1 The empirical inversion

Stage 1 ran the engine on the public grokked modular-addition checkpoint ($p = 113$, one layer, $N = 12,769$ inputs; the re-implementation used here reproduces its behaviour exactly, 12,769/12,769, and its published Fourier structure: the top embedding frequencies $\{14, 35, 41, 42, 52\}$ carry 99.4% of non-constant embedding norm). The engine satisfied the behavioural criterion on 5/5 declared seeds, converging in the first round, in 18–21 seconds per seed, and on 5/5 seeds of each pre-registered ablation (no worked examples; trigonometric primitives removed). The mechanistic criterion failed on 5/5: every recovered program is a one-hot lookup of $(a + b) \bmod 113$, whose logits have the flat spectrum of a delta function (median 51 active Fourier modes against the network's 5).

To locate the failure, the published mechanism was scored directly, written

$$z_c \;=\; \sum_{k \in \{14,35,41,42,52\}} \cos\big(2\pi k\,(a + b - c)/113\big)$$

in the candidate language, with frequencies taken from the network's own embedding spectrum rather than fitted. Table 1 shows the result: the mechanism fits the output distributions marginally *better* and loses the posterior by 9.525 nats, because it costs 190.5 more bits of description and the likelihood cannot repay them. The realised likelihood gap across all 12,769 inputs is 9.46×10^{-5} nats.

4.2 Why no distributional rule can do better

The cause is not the prior's coefficient but the channel. Write $\bar{\varepsilon} = \frac{1}{N}\sum_x (1 - \max_c q_x(c))$ for the network's mean leaked mass; on the grokked network $\bar{\varepsilon} = 2.40 \times 10^{-7}$.

Proposition 1 (Distributional indistinguishability)**.** *Let π be argmax-correct on every input, with logit margin $\gamma = \min_x \big(z^\pi_{x,a_x} - \max_{c \neq a_x} z^\pi_{x,c}\big) > 0$ and spread $R = \max_x \big(\max_c z^\pi_{x,c} - \min_c z^\pi_{x,c}\big)$. If $\bar{\varepsilon} R < (C-1)\gamma$, then*

$$\inf_{\tau > 0} \overline{\mathrm{KL}}_\tau(\pi) \;\le\; \frac{\bar{\varepsilon} R}{\gamma}\left(1 + \ln \frac{(C-1)\gamma}{\bar{\varepsilon} R}\right) \;=:\; B(\pi).$$

Corollary 1 (The prior decides)**.** *For any two argmax-correct candidates A, B as above, $|\ell_{\mathrm{dist}}(A) - \ell_{\mathrm{dist}}(B)| \le N\big(B(A) + B(B)\big)$. Hence whenever*

$$\lambda \,\big|\mathrm{DL}(A) - \mathrm{DL}(B)\big| \;>\; N\big(B(A) + B(B)\big),$$

the maximum a posteriori choice between them is the shorter program, irrespective of which, if either, matches the network's mechanism.

Proofs are one paragraph each (Appendix A); the content is only that a near-deterministic softmax pays $O(\tau R)$ log-loss on $\bar{\varepsilon}$ of its mass and $O(Ce^{-\tau\gamma})$ on the rest, and the optimal temperature balances them at $O(\bar{\varepsilon}\log(1/\bar{\varepsilon}))$.

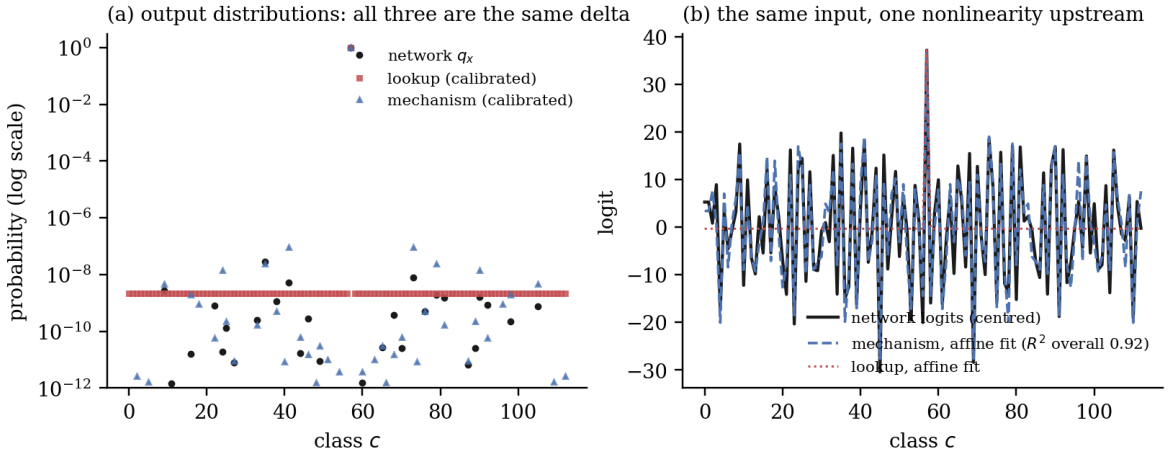


Figure 2: The ceiling and its inverse on one input $(a, b) = (40, 17)$. (a) Output distributions: the network and both calibrated candidates are the same point mass; all sub-argmax structure lies below 10^{-6}. (b) Raw logits: the network's wave is tracked by the mechanism's affine fit and is invisible to the lookup.

Instantiation. The lookup has $(\gamma, R) = (1, 1)$; the mechanism's logit vector is shift invariant in $(a+b-c)$ mod 113, so its margin and spread are exact constants, $\gamma = 2.41$ and $R = 8.771$. The bounds evaluate to $B = 5.02\times10^{-6}$ (lookup; achieved 1.05×10^{-6}) and $B = 1.71\times10^{-5}$ (mechanism; achieved 1.04×10^{-6}), each dominating its achieved value as required, and

$$N\big(B(A)+B(B)\big) = 0.283 \text{ nats} \;<\; \lambda\,\Delta\mathrm{DL} = 9.525 \text{ nats}.$$

Corollary 1 therefore *decides* the selection at the measured parameters: no tuning of the search, the proposer, or the compute budget can make the pre-registered scorer prefer the mechanism. The realised gap (9.46×10^{-5} nats) sits a further three orders below the bound.

Remark 1 (Worst-case mass does not suffice). Replacing $\bar{\varepsilon}$ by the worst-case leaked mass $\varepsilon_{\max}$ ($= 7.34\times10^{-5}$) weakens the right-hand side to 61.851 nats, which exceeds the prior gap: at these parameters the ceiling is a fact about concentration *on average*, and the mean-mass form of Proposition 1 is the one that bites.

Remark 2 (Scope). The corollary conditions on argmax-correct candidates with nondegenerate margins; it says nothing about candidates that disagree with the network, whose likelihoods separate freely, and it is a statement about scoring rules that are functions of output *distributions*. Both restrictions are exactly where Section 5 goes.

5 One nonlinearity upstream

The softmax is what concentrates; the logits are not concentrated. The mechanism predicts a specific wave across all 113 output coordinates; the lookup predicts nothing below the argmax. Figure 2 shows both facts on a single input, and Table 2 quantifies them over the full space: with one global affine calibration per candidate, the mechanism explains 91.6% of the network's logit variance (lookup 8.2%), and on the 112 coordinates below the argmax, where the lookup is constant by construction, the mechanism's mean per-input correlation with the network is 0.956.

Proposition 2 (Logit identifiability)**.** *Model the network's logits as* $z_x = \alpha\, z_x^{\pi} + \beta\mathbf{1} + \sigma\xi_x$ *with* $\xi_x \sim \mathcal{N}(0, I_C)$ *i.i.d., and let* $\mathrm{SSE}(\pi)$ *be the residual sum of squares after the least-squares fit of* (α, β)*. Profiling the likelihood over* (α, β, σ)*, the maximum a posteriori comparison of candidates A (shorter) and B reduces to*

$$\textit{prefer } B \iff \frac{NC}{2}\ln\frac{\mathrm{SSE}(A)}{\mathrm{SSE}(B)} > \lambda\,\Delta\mathrm{DL} \iff \frac{\mathrm{SSE}(B)}{\mathrm{SSE}(A)} < \exp\Big(-\tfrac{2\lambda\Delta\mathrm{DL}}{NC}\Big).$$

candidate	logit variance explained	below-argmax mean r	residual RMS
one-hot lookup	8.2%	0.000 (degenerate)	11.50
published mechanism	91.6%	0.956	3.48

Table 2: Logit-space fits over all 12,769 inputs (1, 442, 897 scalar observations), one affine calibration per candidate.

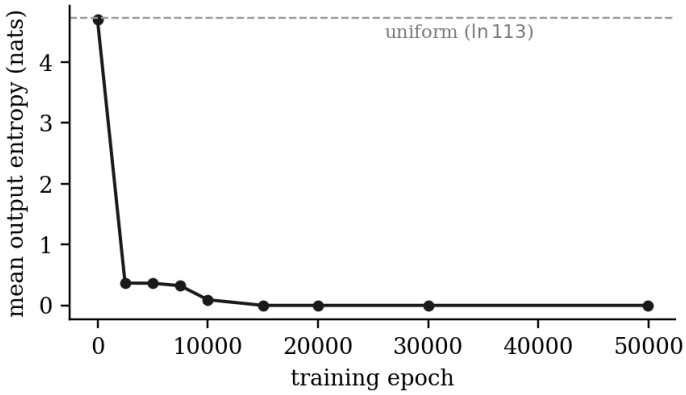


Figure 3: Mean output entropy across training checkpoints (2,000 sampled inputs, seeded). The distributional channel closes early.

At fixed known σ the comparison is instead $\Delta\ell = \big(\mathrm{SSE}(A) - \mathrm{SSE}(B)\big)/(2\sigma^2)$, so the preference for B survives exactly while $\sigma < \sigma^* = \sqrt{\big(\mathrm{SSE}(A) - \mathrm{SSE}(B)\big)/(2\lambda\Delta\mathrm{DL})}$.

Instantiation. Measured, SSE(lookup)/SSE(mechanism) gives ratio 0.092 against a required threshold of $1 - 1.32 \times 10^{-5}$: the profile form prefers the mechanism by 1.72×10^6 nats, five orders beyond the prior gap it must pay. In the fixed-σ form, $\sigma^* = 3015.1$ against an observed mechanism residual of $\hat{\sigma} = 3.48$: the inversion of Section 4.1 reverses, and remains reversed under a 866-fold misspecification of the noise scale. The robustness statement is given first deliberately; the headline nat count is an artefact of $\hat{\sigma}$ and the threshold is not.

Checkpoints are not the escape; logits are. A natural alternative is to score against earlier checkpoints, before the softmax saturates. Measured across training, mean output entropy falls from 4.696 nats at initialisation to 0.366 by epoch 2,500 and to numerical zero by 15,000 (Figure 3): for this network the distributional channel closes long before the published mechanism finishes forming (Nanda et al., 2023), while the logit channel remains open at convergence.

What this does and does not establish. These are *selection* results: given the mechanism among the candidates, a logit-space likelihood prefers it decisively and a distributional likelihood provably cannot. Whether a *search* equipped with the logit likelihood finds the mechanism unprompted is a distinct, pre-registrable question that this paper leaves open; the repository is instrumented to answer it.

6 An exactly graded program-recovery benchmark

Stage 1 is contaminated by construction: the proposer has read the literature that solved its target, so it tests automation, not discovery (its worth as a test is correspondingly limited, and Section 7 shows a trivial baseline ties it). The uncontaminated test is a suite of targets that exist in no corpus.

	small (3.2k)	medium (24k)	large (182k)	all
engine	5/20	3/20	3/20	11/60
B1 random search	1/20	0/20	0/20	1/60
B2 one-shot frontier call	2/20	0/20	0/20	2/60
B3 best-of-n with feedback	5/20	0/20	0/20	5/60
engine, union over both seeds	3/10	3/10	2/10	8/30 targets

Table 3: Recovery on the benchmark: 30 targets × 2 declared seeds, with per-run matched baselines. Wilson 95% intervals for the engine's tier rates: small $[11, 47]$%, medium and large $[5, 36]$%.

Construction. Thirty evaluation programs are drawn from a seeded grammar over RASP (maps, sequence maps, select-aggregate compositions, selector width), rejected if trivial, behaviourally duplicated, or failing compilation, and compiled to transformer weights with Tracr (Lindner et al., 2023); the generation seed, grammar version, and a 78% rejection rate are published. Three size tiers of ten targets each span 2.0 layers and 3,219 mean parameters (small) to 7.0 layers and 182,457 (large), a 57-fold range; every compiled model matches its ground-truth program exactly on the full input space. Three development targets, generated under a disjoint seed, absorbed all tuning.

Grading. Primary: recovery per Definition 4, executed over the full input space (243 to 15,625 points). Secondary: the search/selection decomposition of Definition 4, which localises failures. Third: calibration of stated confidence against empirical recovery (rank correlation with bootstrap confidence intervals, and expected calibration error), pre-registered as kill condition K3. Each engine run is matched, per target and seed, by three baselines under budget parity: **B1**, typed random search over the same primitives with the same number of candidate evaluations (measured sandbox rejection rate 0.000, and the Stage 1 reference program is reachable, appearing verbatim within 21k draws); **B2**, a single strongest-model call shown the same worked examples; **B3**, best-of-n with error feedback and the same number of proposer calls, intended to isolate whether the population machinery earns anything beyond iterated prompting. B3 is not fully matched: the engine's proposer is also told why recent candidates were rejected unscored, and B3's is not (Section 8).

7 Results on the benchmark

7.1 Recovery

Table 3 is the paper's split verdict in one place. Against the pre-registered bar the engine *fails*: 25% at the smallest tier where 50% was demanded, 15% at the largest where 25% was. The gated follow-on stage of the programme therefore did not run. Against its baselines the engine is alone: beyond the smallest tier, random search, a frontier one-shot, and best-of-n with feedback recover *nothing* (the last comparison carries the feedback asymmetry noted in Section 8), while the engine's rate declines only gently ($25 \to 15 \to 15$%) across the 57-fold size range; kill condition K2 (collapse with scale) does not fire. On Stage 1, by contrast, B3 ties the engine at 5/5 and every seed converged in round one: the population machinery is inert where the proposer already knows the answer and is load-bearing exactly where it does not.

Search versus selection. Of the 49 failed runs, 48 are search failures and exactly one is a selection failure: the archive contained an exactly equivalent program whose extra description length the likelihood could not repay on a small input space. The proposal channel, not the scoring rule, is the binding constraint, and the one selection failure is the finite-sample shadow of Corollary 1.

Seeds decorrelate at the boundary. The union over the two declared seeds reaches $8/30$ targets (27%; Wilson 95% $[14, 44]$%): 3 targets at $2/2$, 5 at $1/2$, 22 at $0/2$, with $\Pr(\text{both} \mid \text{any}) = 0.375$ and all three medium-tier successes at 1-of-2. Among the 22 never-solved targets, 7 reached $\geq 90\%$ behavioural agreement at their best (one at 97.6%). Reliability is therefore partially purchasable with retries at a graded competence boundary; this analysis, and any retry extrapolation, is exploratory.

7.2 Calibration

Stated confidence took 48 distinct values across the 60 runs (s.d. 0.26) and is informative: Spearman $\rho = 0.462$ against recovery, $p = 2.06 \times 10^{-4}$, bootstrap 95% CI $[0.251, 0.629]$; kill condition K3 does not fire, against the study's own written prediction (the lab notes record the expectation that it would, for a reason that generalised the wrong stage). It is simultaneously badly overconfident: ECE = 0.629, with the top-bin runs stating ≈ 0.97 and recovering 24% of the time. The pre-registered measure is a relative mass and saturates whenever one candidate dominates; the secondary Good–Turing-adjusted measure, declared in advance, performs *worse* on rank ($\rho = 0.218$, $p = 0.095$) and is reported losing rather than silently dropped. The ranking is usable; the probabilities are not.

8 Limitations

- **Compiled, not trained, targets.** Tracr models are exact and free of superposition; recovery rates here are an upper bound on the trained-model case, and a trained tier with verified ground truth (grokked families; InterpBench-style targets (Gupta et al., 2024)) is the required next build before any claim leaves this regime.
- **The ceiling's empirical scope.** Proposition 1 is general, but its decisive instantiation is one network; the logit-space inversion is demonstrated on the same network. The constants $(\bar{\varepsilon}, \gamma, R)$ are measured, not assumed, and will differ elsewhere.
- **Selection, not discovery.** No experiment here shows a search *finding* mechanism under a logit likelihood; what is shown is only that the likelihood then prefers it. The discovery experiment is specified and unrun.
- **No circuit-discovery comparisons.** ACDC-line methods output component subgraphs, not programs; running them under a bridging metric is future work, and Table 3 makes no claim about them.
- **One proposer family; prompt-feedback asymmetry.** All search used one model family, and error feedback to the proposer reports argmax disagreements only, which handicaps any future logit-likelihood search until the prompt protocol is re-registered.
- **Small n everywhere.** Twenty runs per tier; intervals are wide, and the union analysis doubly so.
- **Instrument defects occurred.** Four reached data; two flattered the method: a one-shot baseline whose output was truncated, and a random-search baseline that would have scored zero by construction. All are ledgered in Appendix C, and the corrected runs are the only ones reported.
- **An unmatched baseline.** The fix for defect 3 in Appendix C gave the engine's proposer the reasons recent candidates were rejected; B3 never received it. Every Stage 2 comparison with B3 therefore favours the engine in one respect beyond the population machinery, and B3's zero recoveries beyond the smallest tier do not isolate the population's contribution. A matched re-run of B3 is required before that claim can be made.

9 Conclusion

Behavioural distillation of a converged network provably cannot pay for mechanism: the output distributions of a confident model bound the distinguishing evidence below any honest description-length gap, and the prior then prefers the account that has no mechanism in it. The evidence was never destroyed; it sits one nonlinearity upstream, where a Gaussian likelihood on logits reverses the preference with orders of magnitude to spare. On the first exactly graded program-recovery benchmark, an engine built on these scoring principles fails the reliability bar fixed in advance and is at the same time the only method tested that recovers anything beyond toy scale, knowing more than it can prove and claiming more than it knows in a way that can now be measured. What remains open is whether search can be made to follow where selection now points.

Reproducibility

Code, data and logs are at **[repository URL]**. Every number in this paper regenerates from logged artefacts by a single offline command, which also verifies the propositions' instantiations against the measured constants, replays a recorded search bit-identically (nine fields, including the full source of the recovered program), and rebuilds every figure. The repository ships the sealed pre-registration, its one amendment with the forcing algebra, the complete laboratory notebook including all four instrument defects, and per-call API logs for every run, reported and excluded.

References

A. Conmy, A. Mavor-Parker, A. Lynch, S. Heimersheim, and A. Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems 36*, 2023.

M. Cranmer, A. Sanchez-Gonzalez, P. Battaglia, R. Xu, K. Cranmer, D. Spergel, and S. Ho. Discovering symbolic models from deep learning with inductive biases. In *Advances in Neural Information Processing Systems 33*, 2020.

P. Del Moral, A. Doucet, and A. Jasra. Sequential Monte Carlo samplers. *Journal of the Royal Statistical Society B*, 68(3):411–436, 2006.

N. Elhage, N. Nanda, C. Olsson, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. https://transformer-circuits.pub/2021/framework.

K. Ellis, C. Wong, M. Nye, M. Sablé-Meyer, L. Morales, L. Hewitt, L. Cary, A. Solar-Lezama, and J. B. Tenenbaum. DreamCoder: Bootstrapping inductive program synthesis with wake-sleep library learning. In *Proceedings of PLDI*, pages 835–850, 2021.

A. Geiger, H. Lu, T. Icard, and C. Potts. Causal abstractions of neural networks. In *Advances in Neural Information Processing Systems 34*, 2021.

A. Geiger, D. Ibeling, A. Zur, et al. Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 26, 2025.

I. J. Good. The population frequencies of species and the estimation of population parameters. *Biometrika*, 40(3–4):237–264, 1953.

P. D. Grünwald. *The Minimum Description Length Principle*. MIT Press, 2007.

R. Gupta, I. Arcuschin, T. Kwa, and A. Garriga-Alonso. InterpBench: Semi-synthetic transformers for evaluating mechanistic interpretability techniques. In *Advances in Neural Information Processing Systems 37, Datasets and Benchmarks*, 2024.

G. Hinton, O. Vinyals, and J. Dean. Distilling the knowledge in a neural network. *arXiv:1503.02531*, 2015.

D. Lindner, J. Kramár, S. Farquhar, M. Rahtz, T. McGrath, and V. Mikulik. Tracr: Compiled transformers as a laboratory for interpretability. In *Advances in Neural Information Processing Systems 36*, 2023.

N. Nanda, L. Chan, T. Lieberum, J. Smith, and J. Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations*, 2023.

nostalgebraist. Interpreting GPT: the logit lens. *LessWrong*, 2020.

C. Olah, N. Cammarata, L. Schubert, G. Goh, M. Petrov, and S. Carter. Zoom in: An introduction to circuits. *Distill*, 2020.

A. Power, Y. Burda, H. Edwards, I. Babuschkin, and V. Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv:2201.02177*, 2022.

J. Rissanen. Modeling by shortest data description. *Automatica*, 14(5):465–471, 1978.

R. J. Solomonoff. A formal theory of inductive inference, parts I and II. *Information and Control*, 7: 1–22, 224–254, 1964.

G. Weiss, Y. Goldberg, and E. Yahav. Thinking like transformers. In *Proceedings of ICML*, 2021.

A Proofs

Proof of Proposition 1. Fix x, write $a = a_x, q = q_x, z = z_x^\pi, p = \mathrm{sm}(\tau z)$. Since $H(q) \geq 0$, $\mathrm{KL}(q\|p) \leq -\sum_c q_c \ln p_c$. For the argmax coordinate, $-\ln p_a = \ln\big(1 + \sum_{c\neq a} e^{\tau(z_c - z_a)}\big) \leq \ln\big(1 + (C-1)e^{-\tau\gamma}\big) \leq (C-1)e^{-\tau\gamma}$, using $z_a - z_c \geq \gamma$ for $c \neq a$ and $\ln(1+u) \leq u$. For $c \neq a$, $-\ln p_c = \tau(z_a - z_c) + \ln\big(1 + \sum_{c'\neq a} e^{\tau(z_{c'} - z_a)}\big) \leq \tau R + (C-1)e^{-\tau\gamma}$, since $z_a - z_c \leq R$. Therefore $-\sum_c q_c \ln p_c \leq q_a(C-1)e^{-\tau\gamma} + (1-q_a)\big(\tau R + (C-1)e^{-\tau\gamma}\big) = (C-1)e^{-\tau\gamma} + (1-q_a)\tau R$. Averaging over x gives $\overline{\mathrm{KL}}_\tau \leq (C-1)e^{-\tau\gamma} + \bar{\varepsilon}\,\tau R$. The choice $\tau^* = \frac{1}{\gamma}\ln\frac{(C-1)\gamma}{\bar{\varepsilon}R}$, positive under the stated condition, yields $(C-1)e^{-\tau^*\gamma} = \bar{\varepsilon}R/\gamma$ and the bound. □

Proof of Corollary 1. $\ell_{\mathrm{dist}}(\pi) = -N \inf_\tau \overline{\mathrm{KL}}_\tau(\pi) \in [-NB(\pi), 0]$ for each argmax-correct candidate, since KL is nonnegative and Proposition 1 bounds the infimum. Two numbers lying in $[-NB(A), 0]$ and $[-NB(B), 0]$ differ by at most max of the two widths, hence by at most the sum. If the prior gap $\lambda|\mathrm{DL}(A) - \mathrm{DL}(B)|$ exceeds that sum, the log-posterior difference has the sign of the prior term. □

Proof of Proposition 2. For fixed π the model is linear regression of the NC stacked network logits on the candidate's, with Gaussian errors. Maximising over (α, β) gives the least-squares residual $\mathrm{SSE}(\pi)$; maximising over σ gives $\hat{\sigma}^2 = \mathrm{SSE}/NC$ and maximised log-likelihood $-\frac{NC}{2}\big(1 + \ln(2\pi\mathrm{SSE}/NC)\big)$. The difference of maximised log-likelihoods between candidates is $\frac{NC}{2}\ln(\mathrm{SSE}(A)/\mathrm{SSE}(B))$, and adding the prior terms $-\lambda\mathrm{DL}(\cdot)$ gives the first display; the second is algebra. At fixed σ the log-likelihood difference is $(\mathrm{SSE}(A) - \mathrm{SSE}(B))/(2\sigma^2)$, decreasing in σ, and equals $\lambda\Delta\mathrm{DL}$ exactly at σ^*. □

B Pre-registered criteria and verdicts

Pre-registered rule	Threshold	Verdict
Stage 0, instrumentation (two criteria)	embedding concentration $\geq$ 80%; attention periodicity clause	criterion 1 passes (99.4%); criterion 2 fails; amendment A1
Stage 1 success (S1-A)	exact equivalence, any of 5 seeds	passes, 5/5
Stage 1 mechanism (S1-B, secondary)	periodic-family structural and behavioural test	fails, 0/5
K1	S1-A false on all 5 seeds, none budget-cut	does not fire
Stage 2 pass	small $\geq$ 50% and large $\geq$ 25% and $\rho > 0$ at $p < .05$	**does not pass**
K2	large $\leq$ 5% while small $\geq$ 50%	does not fire
K3	ρ with $p \geq .05$ or bootstrap CI $\ni 0$	does not fire

Table 4: Every pre-registered decision rule and its outcome. No run reached its pre-registered compute or wall-clock cap; wall clock used was about 6 minutes of 48 h declared (Stage 1) and about 2.5 h of 24 h (Stage 2). The Stage 0 amendment is published with the algebra showing the original clause had a hard ceiling of $1 - \text{mean} \approx 0.502$ against a 0.5 threshold.

C Instrument defect ledger

Four defects reached data. For each, the direction is the effect it would have had on the reported comparisons had it gone unfixed, re-derived here from its consequence. (The laboratory notebook's contemporaneous tally named the wrong pair as flattering; the notebook records this correction.)

1. **One-shot baseline truncated** (flattering). Baseline B2 was capped at the engine's per-call output allowance; replies were cut off before any program could be parsed, so the baseline scored zero because of the experimental setup, not its own behaviour. Fixed by giving the single call room to finish; the affected pass was discarded and re-run.
2. **Scalar sampler on sequence targets** (flattering). The random sampler emitted scalar-shaped programs: 0/400 valid on sequence targets. Baseline B1 would have scored zero *by construction*. The same defect disabled the engine's top-up channel, which cut against the engine, but the net effect on the comparison favoured it. Rebuilt (400/400 valid); all affected runs discarded and re-run.
3. **Rejection reasons discarded** (unflattering). 23.8% of early Stage 2 candidates were rejected unscored, mostly for syntax the output format forbids, and the engine never told the proposer why. Wasted budget lowers recovery. Fixed by feeding the reasons back; the final-run invalid rate is 13.4%. The fix was applied to the engine only (Section 8).
4. **Population collapse** (unflattering). A generation with no valid candidates ended the run at round one. Fixed by topping up from the sampler.

Two further defects affected cost accounting only (a shared spending meter read as per-run spend, and an even split of spend across baselines). This paper reports no costs, so neither touches any reported quantity. Stage 1 and Stage 2 were re-run in full under the corrected instruments; the discarded passes are archived, not deleted.

D Reproduction

In the repository at **[repository URL]**, `make reproduce` verifies the frozen experiment configurations against the code, runs the 144-test suite and the end-to-end harness gate, regenerates Stage 0 from the

public checkpoint (SHA-256 pinned), replays a recorded Stage 1 search offline from its own API log and confirms bit-identity on nine fields, recomputes the propositions' instantiations with assertions that every bound dominates its achieved value, and rebuilds all figures and tables. Live re-runs reproduce the protocol but not the digits: the proposer API is not seedable, which is why replay, not re-execution, is the reproducibility claim. Declared seeds: Stage 1 $\{11, 23, 47, 101, 211\}$; Stage 2 target generation 20260828, run seeds $\{11, 23\}$; development 909.